

Document Classification - AI

Executive Summary:-

The Document Classification Portal is an AI-powered system that automatically reads, understands, and classifies HR documents submitted by Walmart associates. It replaces a manual, error-prone review process with a self-improving ML pipeline that auto-approves confident predictions at zero cost.

The Problem :

Walmart's associate onboarding and offboarding process generates hundreds of millions of documents annually — each requiring accurate classification before HR and compliance workflows can proceed. Vault, Walmart's centralized in-house document storage platform, receives uploads from across the organization, but there is no automated guarantee that uploaded documents are classified correctly. Today, classification is largely manual: associates open documents, read them, assign types — slow, inconsistent, error-prone, and unscalable at Walmart's volume.

Approach	Problem
Manual Classification	Impossible at a scale of 180M+ documents due to the required human effort.
Rule-Based Systems	Brittle and require frequent maintenance; they break as document types and formats evolve.
100% LLM Classification	Costs approximately \$0.03–\$0.10 per document , making it prohibitively expensive at large scale.

Neither learns from corrections. Neither improves over time. Neither gives associates transparency into why a document was classified a certain way.

The opportunity:

Build a system that starts smart using LLMs, continuously learns from human feedback, and progressively shifts classification to a free local ML model — eliminating per-document cost while improving accuracy. Every human correction becomes training data, not wasted effort.

Doc Mind AI introduces a three-tier intelligent classification pipeline with a self-learning feedback loop:

Tier 1 — Local ML (Free, ~1 second)

Documents are embedded using nomic-embed-text (768-dim semantic vectors) and classified by Tribuo LogisticRegression trained on approved documents. If confidence 80%, the document is auto-approved instantly at \$0 cost. No LLM call. No waiting.

Tier 2 — LLM Fallback (Gemini, ~\$0.03/doc)

When ML confidence is below threshold — or during cold start — Gemini via Walmart's LLM Gateway classifies the document with reasoning, surfaced to a human reviewer.

Tier 3 — Human Review Portal (the flywheel)

The business team reviews predictions in a browser UI with full document preview, confidence scores, ML reasoning, and one-click approve/reject. Every approval feeds directly back into model training.

1.Where We Started: Document Azure OCR LLM (Gemini/GPT) Predicted Type Human Reviews . This worked — but had serious problems:

Every single document required a paid LLM API call (~\$0.01–\$0.05 per doc) . No learning — same cost and same time, day after day, forever . At scale, cost explodes.

2.Can We Do Better?

What if a cheaper, faster local model could handle the easy cases?Instead of always calling the expensive LLM, use a two-step approach:

Try a local ML model first (fast, free)

Only call the LLM if the ML model isn't confident.

Refer : **Doc-Mind-AI confluence page (The Hackathon Project)**

Classification Method	Correct	Wrong	Accuracy	Total Classified
ML Classified	64	86	42.67%	150
LLM Classified	69	5	93.24%	74
Pending (Unknown/Under Review)	–	–	–	3
Failed (no type assigned)	–	–	–	6
Grand Total	133	91	59.37%	224 classified

Initially, for feature extraction and model training, we used TF-IDF. However, we found that most documents were biased toward the NDA class, which affected the model's performance. To address this limitation, we transitioned to **Ollama**- models, which provided better contextual understanding and improved classification accuracy.

What is Ollama?

Ollama is a tool that lets you run AI models locally on your laptop — for free, with no internet required.

A local server that hosts AI models. You send it text, it returns AI output

Your App POST <http://localhost:11434/api/embeddings> Ollama AI output

Models that I pulled : smolLM 135M , phi3 3.8B , nomic-embed-text 137M

1. SmolLM— The First Attempt:

The first local model tried was SmolLM2:135m — a tiny 135-million parameter language model. How it worked:

a. Ask SmolLM2 14 yes/no questions about the document:

"Does this document mention salary?" , "Does this document have a signature line?" , "Does this contain legal language?" etc

b. Collect 14 binary answers (0 or 1) feed into Tribuo ML classifier

Tribuo predicts document type from those 14 features

The problem: 14 yes/no features = very coarse signal

An NDA and a Background Check both answer "yes" to legal language look identical to the model

2 . The Switch to nomic-embed-text:

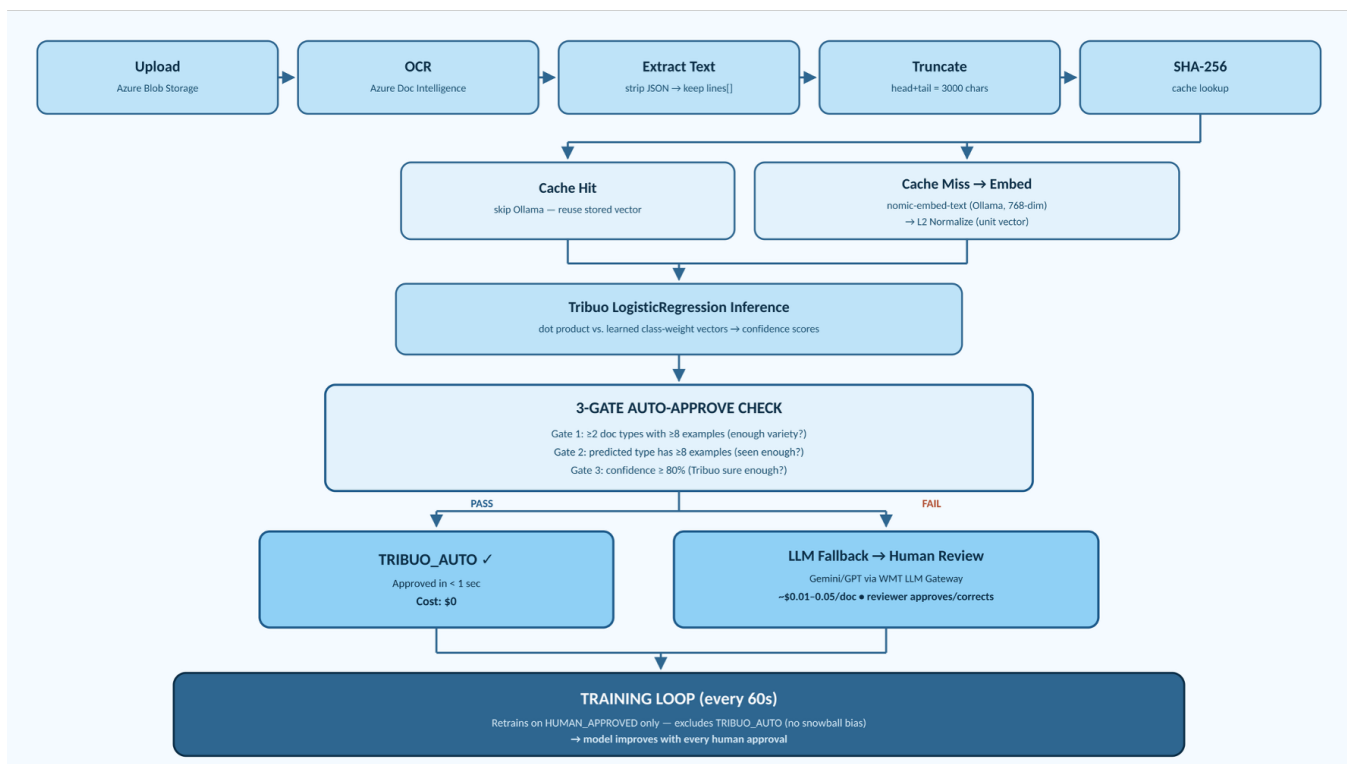
it converts the entire document text into 768 numbers — a mathematical fingerprint of the document's meaning.

"CONFIDENTIALITY AGREEMENT between Walmart Inc. and..."

[0.0055, -0.107, 0.832, 0.041, -0.293, ...] 768 numbers

These 768 numbers represent where the document sits in semantic space. Documents with similar meaning produce similar vectors. Documents that are different produce vectors pointing in different directions.

The WorkFlow:



Accuracy :-

Metric	Value
Total Documents	582
Approved	560
Under Review	12
Rejected	0
Auto Approved by ML	364
ML Accuracy	~95%

Till Now Best Accuracy

The WCNP Deployment issue :As this ollama is not approved by walmart , im running it locally but cant deploy it , so we are looking for something that works in same way but also approved by walmart.

then the Element-AI came into picture.

3.Try to replace Ollama with Element AI:

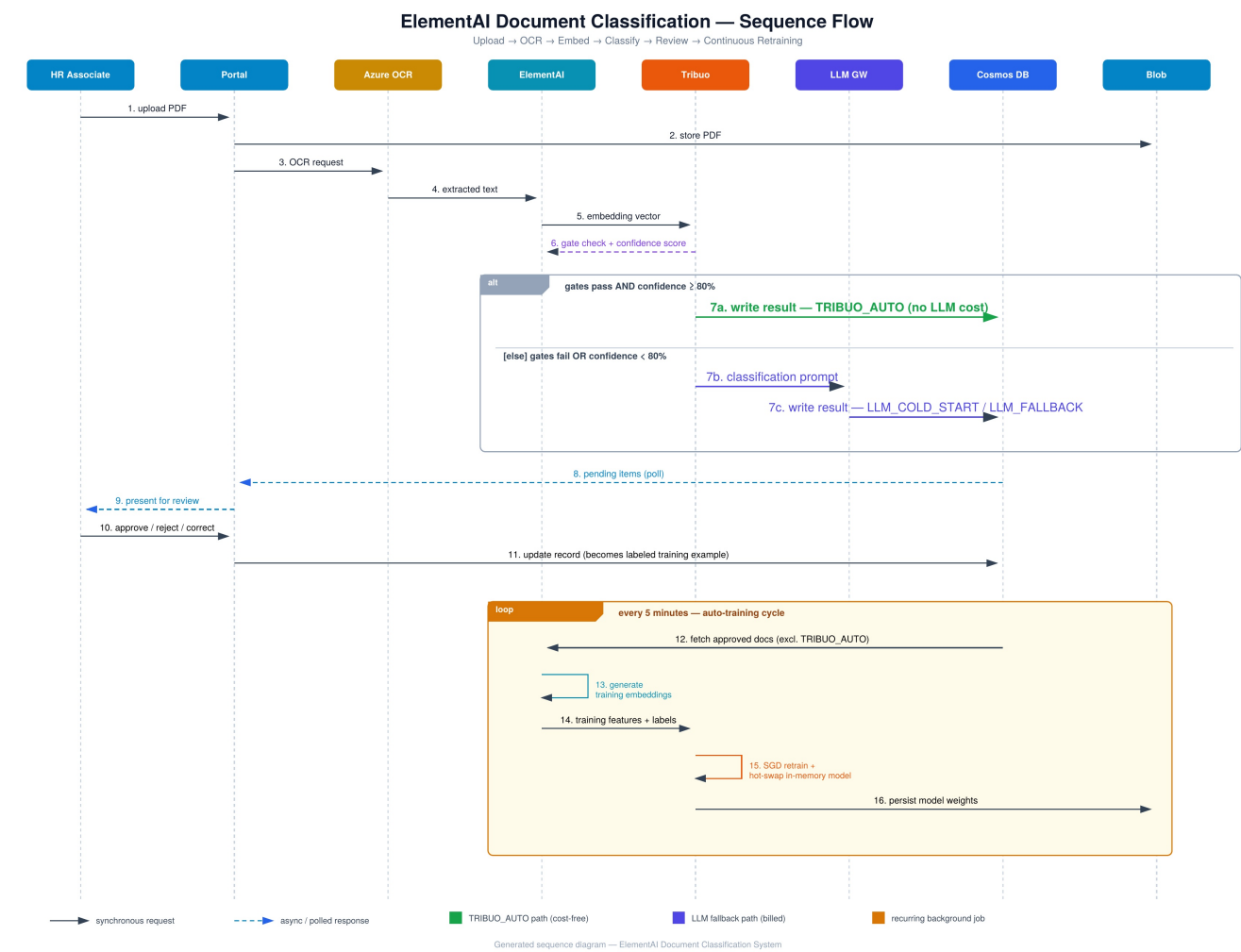
ElementAI is a Walmart internal AI/ML service used for embeddings. ElementAI provides:
Text embeddings accessed via the WMT LLM Gateway (wmtllmgateway.stage.walmart.com)
Used for converting document text into vector representations for semantic understanding

Comparison of this element ai and this ollama : -

Factor	ElementAI (WMT Gateway)	Ollama
Price per 1K Tokens	\$0.01	\$0.00
Hosting	Walmart-managed SaaS (WMT Gateway)	Self-hosted (your own hardware)
Latency	Network call over WMT Gateway	Local inference only
Model Updates	Managed by Walmart	User-managed updates

API Calls	HTTP requests to <code>wmtllmgateway.stage.walmart.com</code>	Local process (no HTTP calls)

ElementAI sequence diagram :



The Accuracy is Good can say 90-95%,

Results after integration with element ai :

ML bypass Rate: 76%

Can be deployable - Only the Cost issue is there

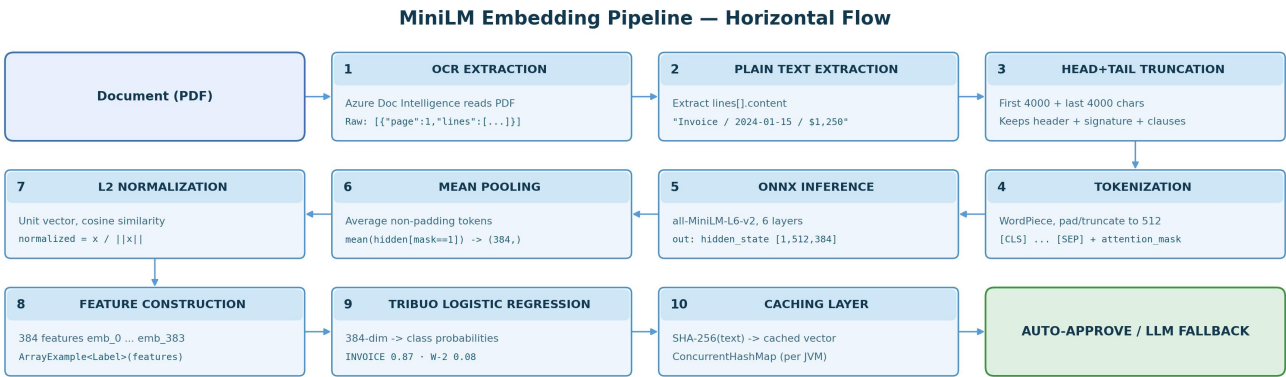
DocMind AI — Element AI Portal text-embedding-3-small · 1536-dim Embeddings · Tribuo ML · WMT LLM Gateway					
Documents	Dashboard	Accuracy	Cost Savings	Refresh	Clear All
TOTAL 788	UNDER REVIEW 19	APPROVED 755	REJECTED 2	AUTO APPROVED BY ML 594	LLM CLASSIFIED 182
All Under Review Approved Rejected Search by document name or type... Rows: 25					
DOCUMENT NAME	FINAL DOCUMENT TYPE	CLASSIFICATION METHOD	REVIEW STATUS	ACTIONS	
638542728_candidate_pii.pdf	Associate Information Form	LLM Fallback	Under Review	Review	
638542728_hsa_fsa_memo.pdf	Benefits and SMM Acknowledgment Forms	LLM Fallback	Under Review	Review	
638542728_emp_letter_hourly.pdf	UNKNOWN	-	Failed	Review	
638542728_offer_letter_update.pdf	Job Offer	Auto Approved by ML	Approved ✓	Review	
638542728_offer_wm_senior.pdf	Job Offer	Auto Approved by ML	Approved ✓	Review	
638542728_offer_wm_staff.pdf	Job Offer	Auto Approved by ML	Approved ✓	Review	
638542728_offer_vizio.pdf	Job Offer	Auto Approved by ML	Approved ✓	Review	
638542728_nda_template.pdf	NonDisclosure Agreement	Auto Approved by ML	Approved ✓	Review	
638542728_pii_cert.pdf	Proprietary Information & Inventions Agreement (PIIA)	Auto Approved by ML	Approved ✓	Review	

Since this Element AI is costly again we are looking for walmart approved open source models -

4 . MiniLM6v2 -> Hugging Face Models

- MiniLM Specifications
- Created by: Sentence-Transformers (Hugging Face)
- Type: BERT-based sentence embedding model
- Parameters: 22 Million (22M)
- Download size: ~44 MB (uncompressed)
- Output dimensions: 384-dimensional vectors
- Max input tokens: 512 (BERT limit)
- Speed: ~200-300ms per document (CPU)
- Architecture: 6 Transformer layers + attention heads

The Workflow :

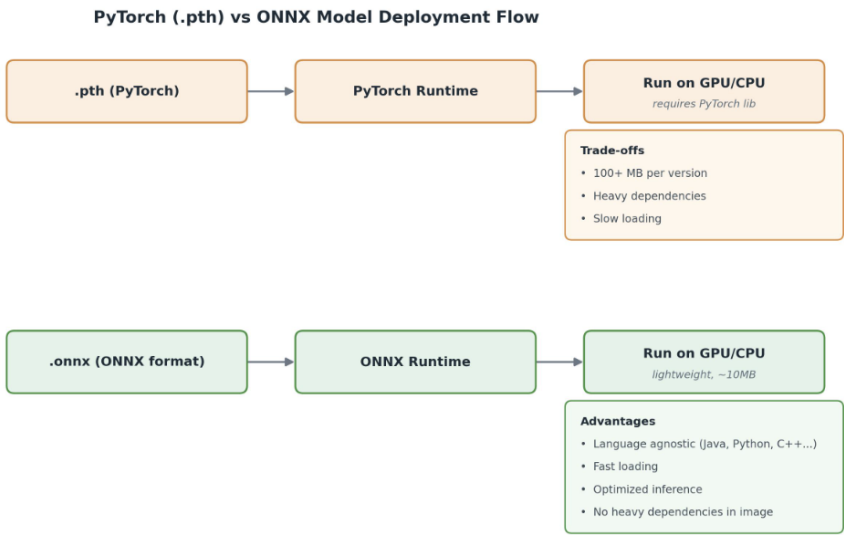


Accuracy : ML Bypass Rate : 70%

DocMind AI — HuggingFace Portal all-MiniLM-L6-v2 · 384-dim ONNX · Local · No Embedding API · Tribuo ML					
Documents		Dashboard		Accuracy	
Cost Savings		Refresh		Clear All	
TOTAL 360	UNDER REVIEW 2	APPROVED 346	REJECTED 1	AUTO APPROVED BY ML 251	LLM CLASSIFIED 98
All Under Review Approved Rejected					
Search by document name or type...					
Rows: 25					
DOCUMENT NAME	FINAL DOCUMENT TYPE	CLASSIFICATION METHOD	REVIEW STATUS	ACTIONS	
628143974_background_check.pdf	Criminal Background Check Result	Auto Approved by ML	Approved ✓	Review	
619838730_offer_letter_update.pdf	Job Offer	Auto Approved by ML	Approved ✓	Review	
619838730_powerform.pdf	Associate Information Form	Auto Approved by ML	Approved ✓	Review	
619838730_offer_vizio.pdf	Job Offer	Auto Approved by ML	Approved ✓	Review	
619838730_schedule_a.pdf	Terms of Employment	Auto Approved by ML	Approved ✓	Review	
619838730_offer_wm_staff.pdf	Job Offer	Auto Approved by ML	Approved ✓	Review	
619838730_piia_template.pdf	Proprietary Information & Inventions Agreement (PIIA)	Auto Approved by ML	Approved ✓	Review	
619838730_offer_wm_senior.pdf	Job Offer	Auto Approved by ML	Approved ✓	Review	
619838730_offer_letter.pdf	Job Offer	Auto Approved by ML	Approved ✓	Review	

5. Nomic-embed-text-v1.5

works same as nomic Embed text - the second one.



Property	SmoLM	ElementAI	MiniLm	Nomic-embed-text
Model	smoLM:135m	text-embedding-3-small via WMT gateway	all-MiniLM-L6-v2	nomic-embed-text-v1.5
Dimensions	384	1536	384	768
Model -size	~135MB	Cloud-no download / api call	86	270
Cost per call	\$0	Api cost	\$0	\$0
Max-Tokens	2048	2048	512	8192
Accuracy	Poor	Best	good	Better than Minilm and smoLM
Walmart Artifactory	No	Yes	Yes	No

*Documents for which i have tested - 1. M & A ai gen docs

2.Efile Doc samples

The final approach - that i have chosen is using the MiniLM Model - fast , less MB , easy implementation , approved , accurate.

Repo Link : <https://github01.walmart.com/corehr/docu-mind-ai/tree/MiniLm-Model>

Branch - MiniLm-Model

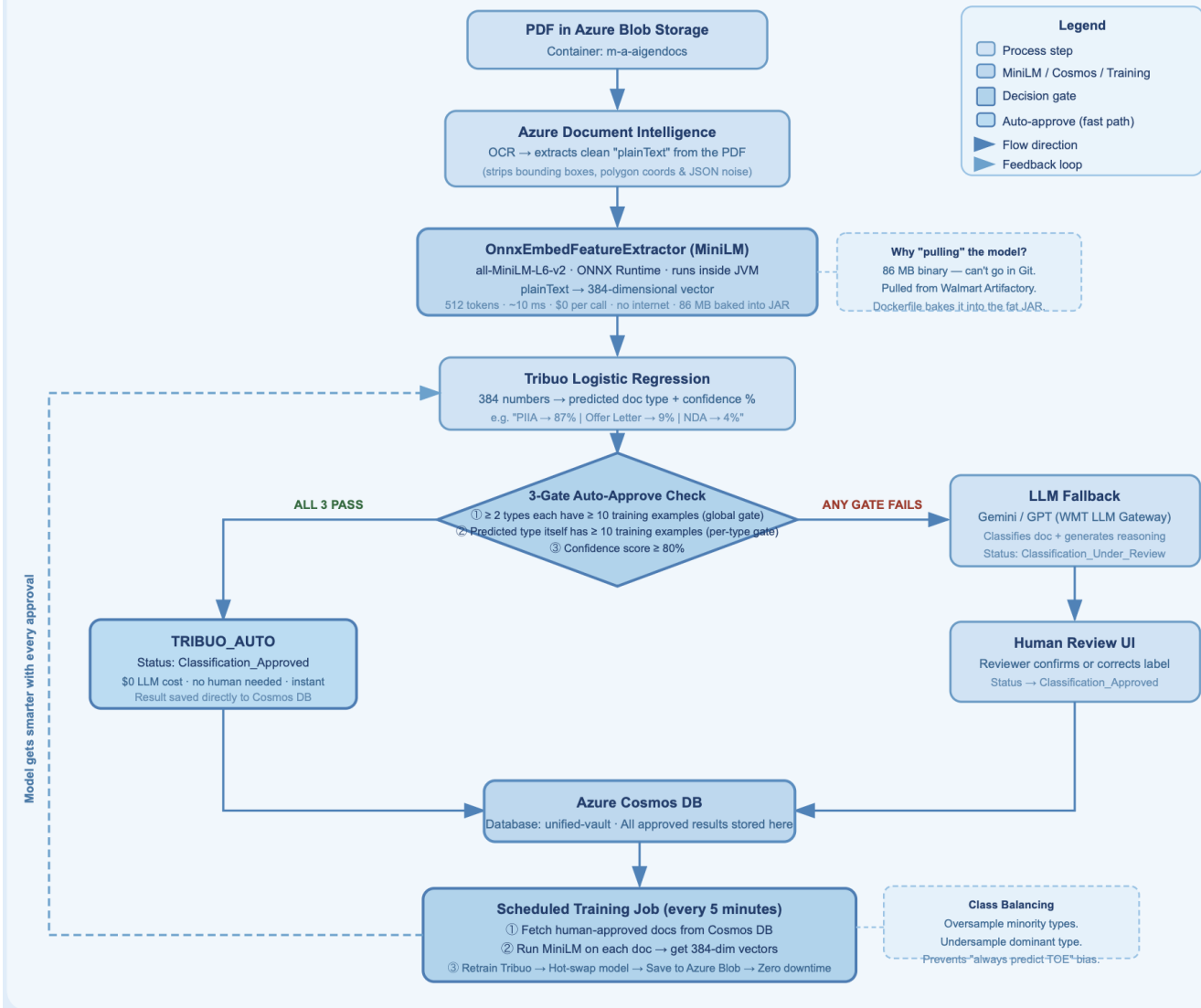
Dev deployed Link :

<https://doc-mind-ai.dev.walmart.com>

The Workflow :

DocMind AI — Document Classification Workflow

How a PDF goes from Azure Blob → auto-classified or human-reviewed, and how the model keeps learning



The Accuracy : -

Metric	ML	LLM
Docs classified	360	209
Correct	~360	102
Wrong	0	107
Accuracy	~90-95%	68.4%
Avg speed	~66 ms	11,998 ms
Bypass rate	63.3%	—
Tokens used	\$0	2,864,007 tokens

DashBoard :

TOTAL
640

UNDER REVIEW
0

APPROVED
598

REJECTED
0

AUTO APPROVED BY ML
388

LLM CLASSIFIED
210

AllUnder ReviewApprovedRejected

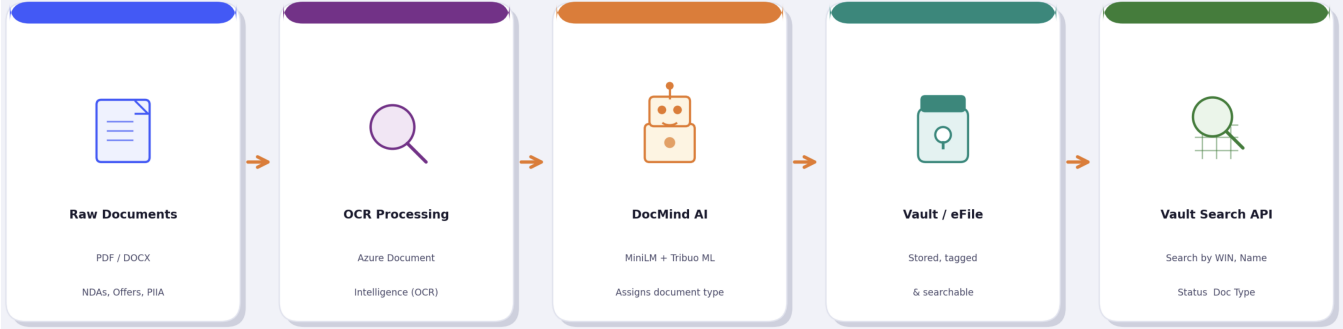
Search by document name or type...

Rows: 25

DOCUMENT NAME	FINAL DOCUMENT TYPE	CLASSIFICATION METHOD	REVIEW STATUS	ACTIONS
100000013_PIIA.pdf	Proprietary Information and Inventorship Agreement	Auto Approved by ML	Approved ✓	Review
100000012_Salary Offer Letter.pdf	Job Offer	Auto Approved by ML	Approved ✓	Review
100000013_I9 Employment Eligibility.pdf	I-9 Form	LLM Fallback	Approved ✓	Review
100000014_CIIA Amendment.pdf	Statement of Confidentiality	Auto Approved by ML	Approved ✓	Review
100000014_Job Change Form_1.pdf	Internal Job Change	Auto Approved by ML	Approved ✓	Review
100000014_Photo Consent Form_1.pdf	General Release for Photograph	Auto Approved by ML	Approved ✓	Review
100000014_Leave of Absence_1.pdf	Leave of Absence	Auto Approved by ML	Approved ✓	Review
100000014_I9 Employment Eligibility.pdf	I-9 Form	Auto Approved by ML	Approved ✓	Review
100000013_Statutory Forms_1.pdf	Beneficiary Forms	Auto Approved by ML	Approved ✓	Review
100000013_Salary Offer Letter.pdf	Job Offer	Auto Approved by ML	Approved ✓	Review
100000013_Project Vista PIIA.pdf	Proprietary Information and Inventorship Agreement	Auto Approved by ML	Approved ✓	Review

Next steps : Vault integration

Vault Integration Path:
Doc Mind AI is architected as the classification intelligence layer for Vault — ensuring every document uploaded to Walmart's centralized storage platform is accurately typed at ingestion, before it is stored or routed for compliance review.



What other things that i explored during my Intern - 1. Rag Embeddings 2. Prompting methods like Few shot , Zero shot.